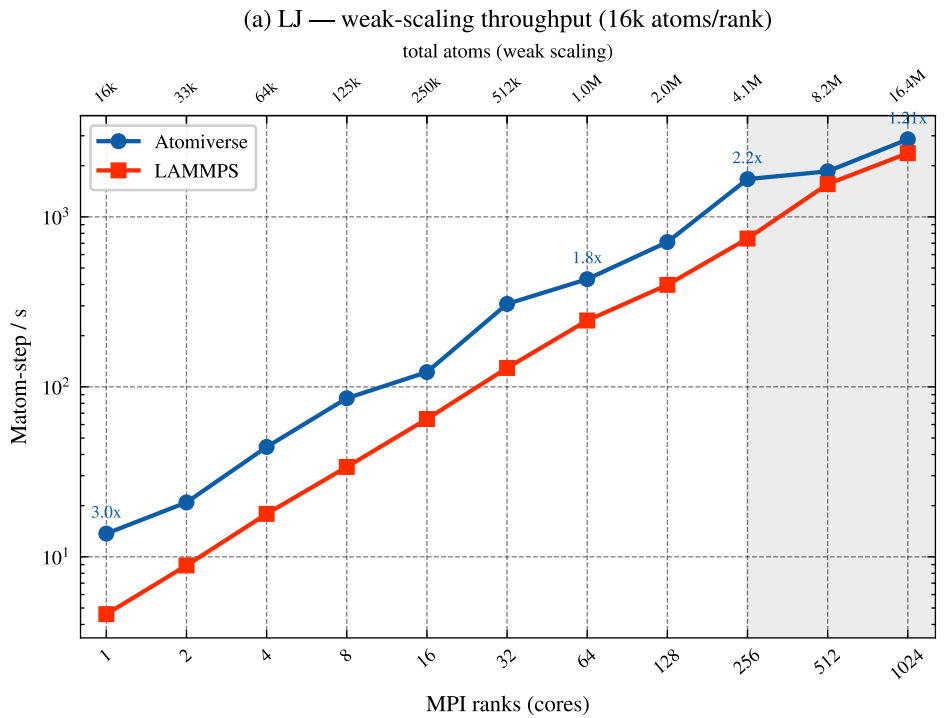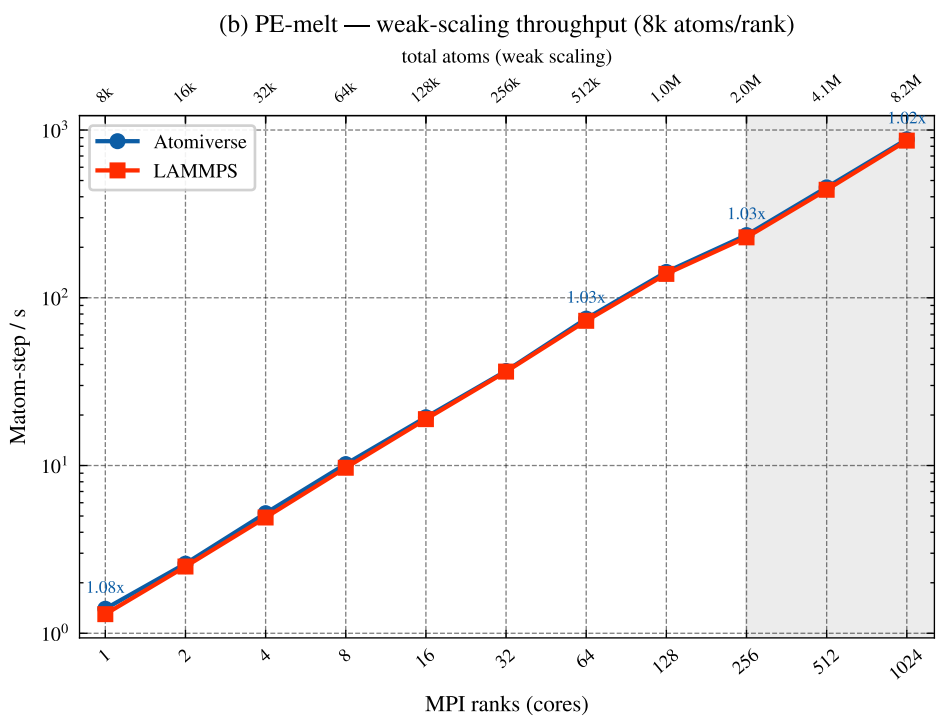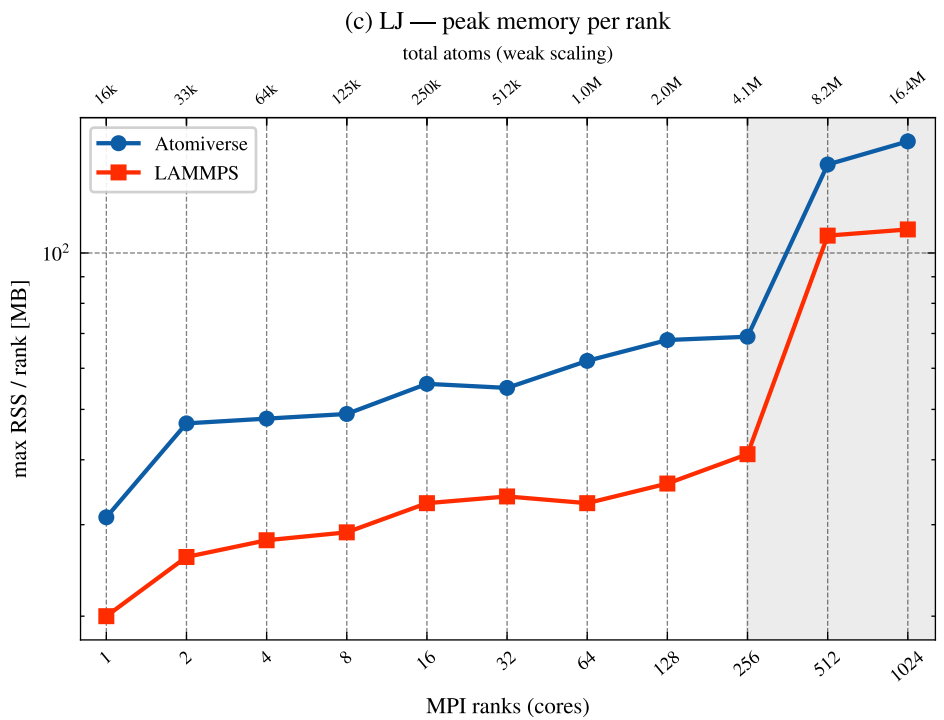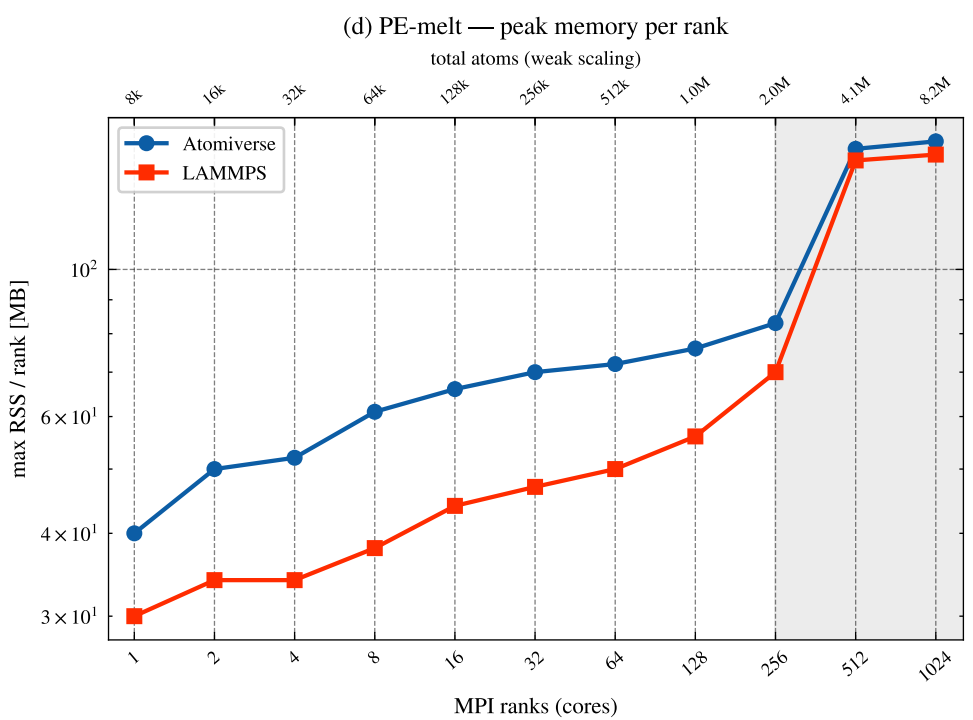Atomiverse vs LAMMPS — CPU multicore weak scaling (EPYC, 1-4 nodes; shaded = cross-node). AV leads throughput at every scale (LJ 1.2-3.0x, PE 1.02-1.06x); memory within ~1.5x.